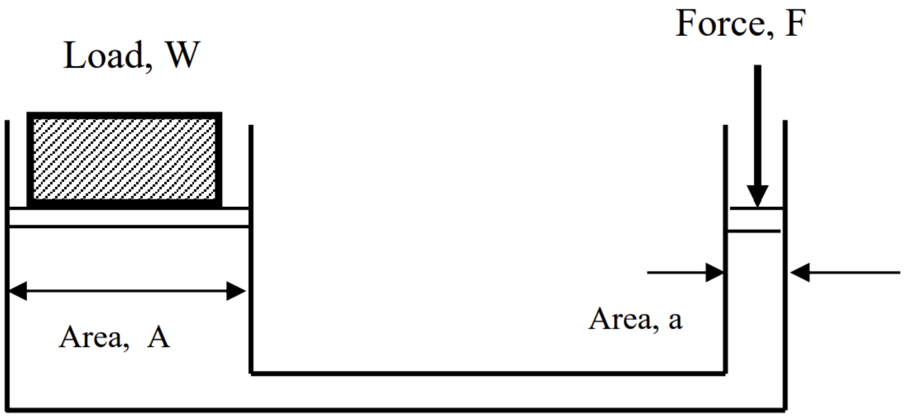
Industrial Hydraulic Systems

The fundamental working principle of hydraulic systems is Pascal's Law, which states that

$$P = \frac{F}{A}$$

In a generic hydraulic system, some applied force is used to do some work. Hence,

$$P_{\text{force}} = P_{\text{load}}$$
$$\frac{F}{a} = \frac{W}{A}$$
$$W = F \times \frac{A}{a}$$



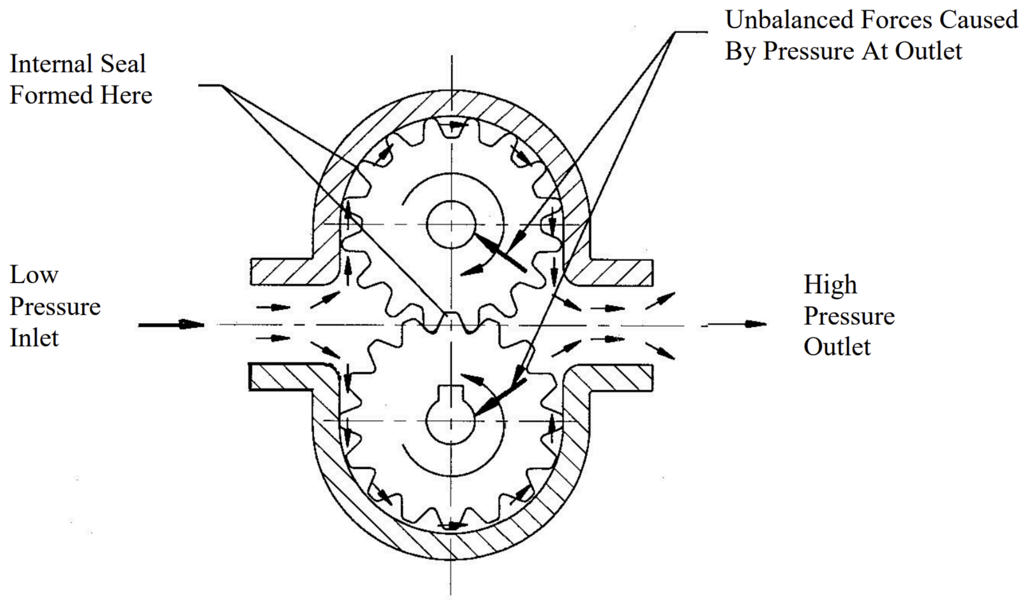
Hence, the applied force is amplified by a factor of $\frac{A}{a}$.

These hydraulic systems are often used for mechanical handling, machine tools, and press work.

In hydraulic systems, there are many different types of components used to power, control and drive hydraulic outputs.

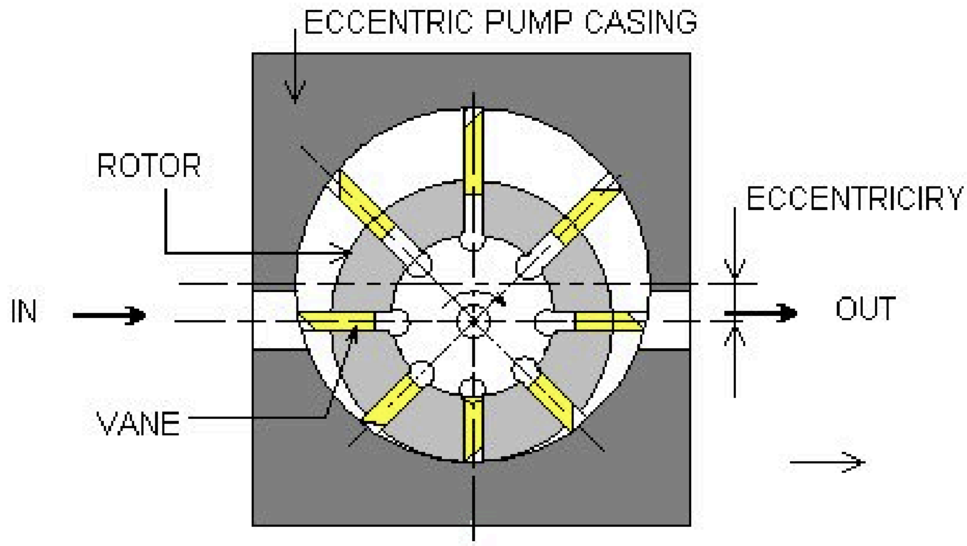
Hydraulic pumps are used to displace fluids to power hydraulic systems.

One type of hydraulic pump is the gear pump:

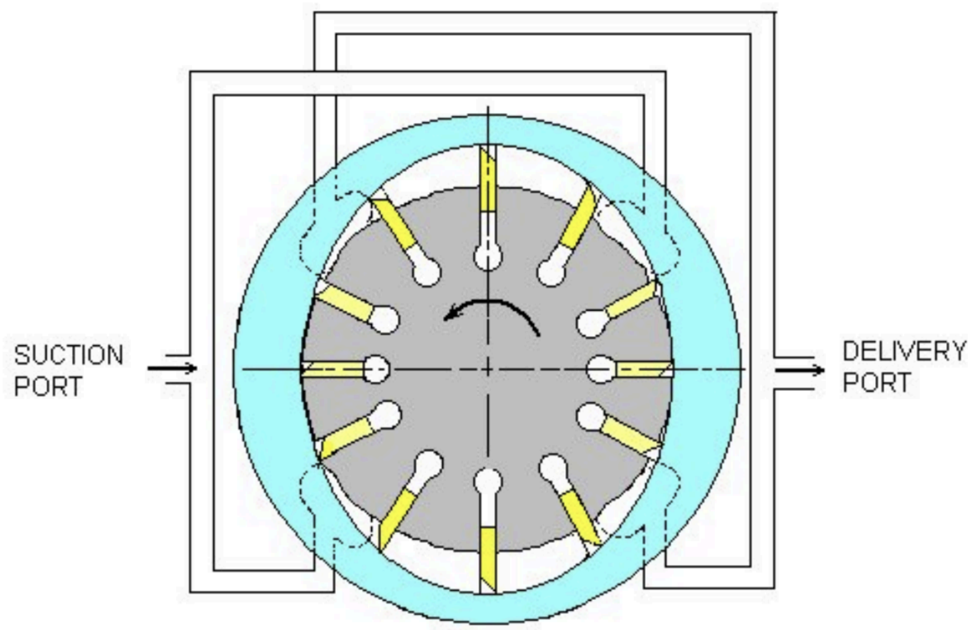


Gear pumps work by having two gears mesh with each other to push fluid around the gears, creating an internal seal between the meshing teeth which forces fluid to flow around the gears to be pressurised. Most gear pumps are fixed displacement, but have their volumetric output range from very low to high volume. However, since the pressure at the outlet creates unbalanced forces, they typically produce low pressure. Gear pumps are also used for pumping medium to high viscosity fluids that are clean, but are durable enough to work with debris, unlike vane or piston pumps

Vane pumps work by having a slotted rotor being driven by a motor with vanes in each slot that move outward against the casing. There are two types of vane pumps, unbalanced and balanced. Unbalanced vane pumps use an eccentric pump casing to push the vanes against the casing, allowing for the fluid to be squeezed out at the outlet. However, the imbalanced design causes unbalanced forces and pressures to be developed, which limits the size of the pump (the flow rate the pump produces)

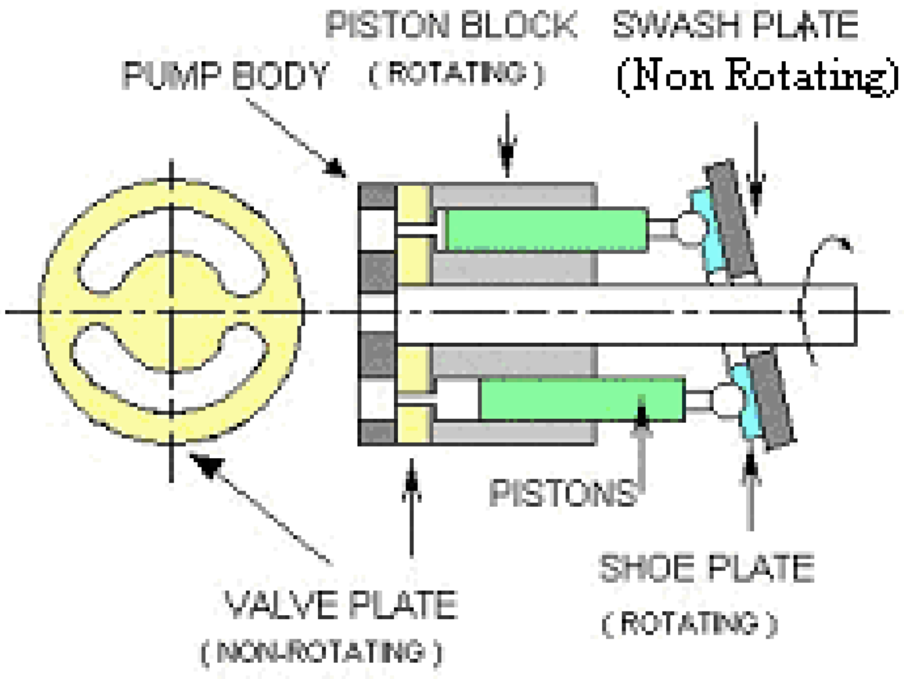


Balanced vane pumps use a symmetric casing and splits the suction port into two, so that each section of the casing does work into pumping the fluid. Similarly, there are two delivery ports in the casing that join together.

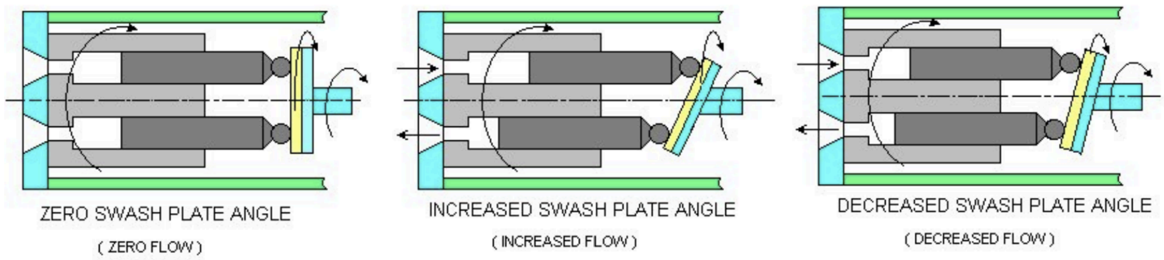


This means that unbalanced pressures in the unbalanced vane pump are cancelled out in the balanced vane pump, allowing for a higher pump size

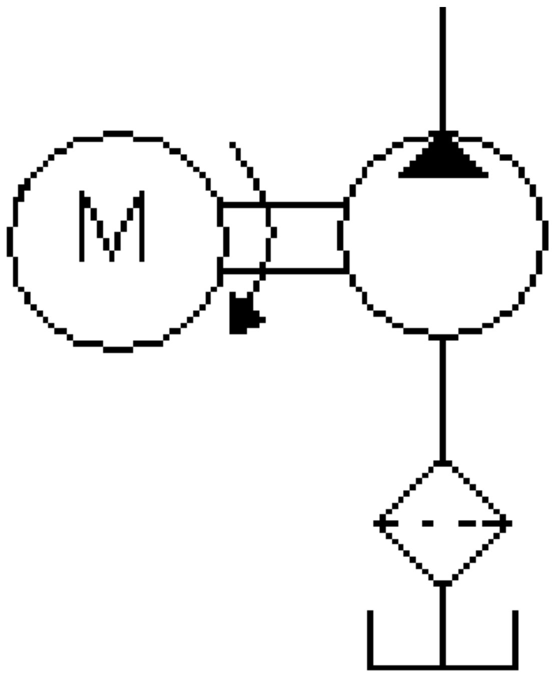
Piston pumps are typically used for high flow rate and high pressure operations, acting similarly to PDPs but with multiple pistons. Typically, instead of using a crank to drive the pistons, a swashplate and rotating piston block is used instead.



As the piston block rotates, the pistons follow the angle of the swashplate, causing the pistons the extend and retract, allowing fluid to be taken in through one slot in the valve plate and ejected through the other after the fluid is compressed by the pistons. Typically, the swash plate angle is fixed, but its angle can be varied to get a variable displacement pump.



In hydraulic circuits, these pumps are part of a power pack, a series of components that produce flowing fluid for other components to use.

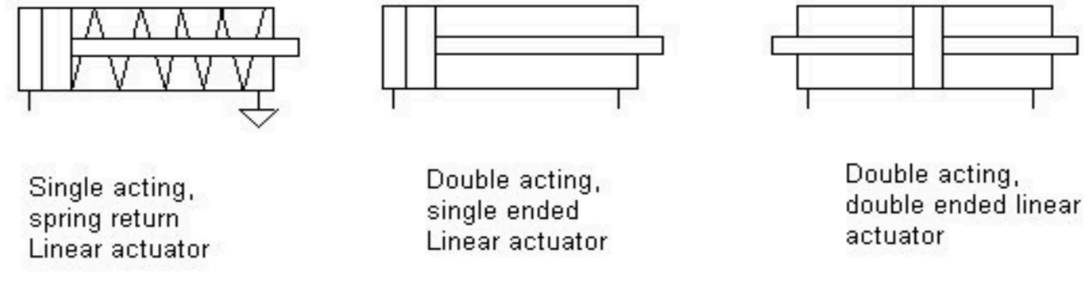


A typical power pack sources fluid from a tank, before being passed through a strainer and into the pump. To power the pump, a motor is connected to the drive shaft. Hence, the series of symbols for the power pack is shown above.

The hydraulic fluid used in hydraulic systems also has specific purposes. Other than transferring hydraulic energy, it helps to lubricate moving parts, seal clearances between parts, and dissipates heat from friction.

Similarly, the tank also has purposes other than storing the hydraulic fluid, like cooling of the fluid by providing a radiating surface (like a heatsink), seperating dirt, fluid and air, providing a filling point, and making up leakages in the system

The output of hydraulic circuits are actuators that are able to do work. The most common type is a hydraulic cylinder, which requires pressure and flow rate to drive the piston forwards or backwards.

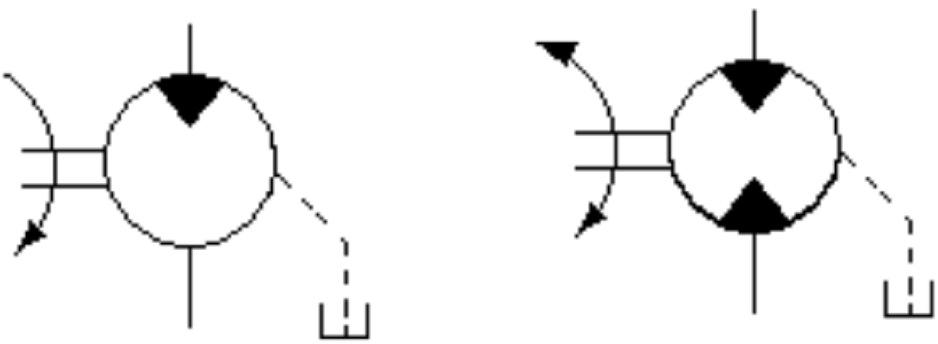


The force and velocity of the piston depends on the pressure and flow rate respectively. Additionally the force and velocity of the forward and return stroke will vary too, since both depend on the area which is not equal on each side.

For a double acting cylinder of piston diameter D, rod diameter d, supplied by a fluid of pressure P and flowrate Q,

$$F_f = P\left(\frac{\pi}{4}D^2\right) \quad F_r = P\left(\frac{\pi}{4}(D^2 - d^2)\right)$$
$$v_f = \frac{Q}{\frac{\pi}{4}D^2} \quad F_r = \frac{Q}{\frac{\pi}{4}(D^2 - d^2)}$$

Another actuator is the hydraulic motor, which converts hydraulic energy into rotational motion. They are typically the same construction as vane pumps, and some pumps can be used directly as hydraulic motors. Similar to cylinders, their rotational speed depends on the flowrate while their torque depends on the pressure.

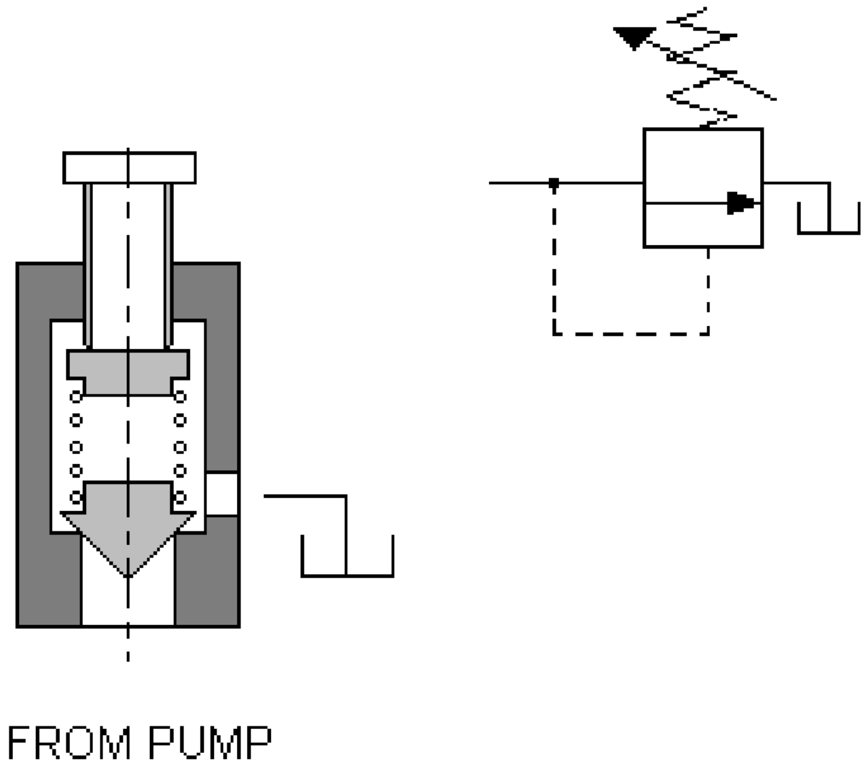


Fixed displacement hydraulic motor
Rotation and flow in a single direction

Fixed displacement hydraulic motor
Rotation and flow in either direction

To control these actuators, we can use pressure valves to control the flow of the the fluid.

One type is the relief valve, which limits the pressure in the main system, protecting components and limiting actuator output. There are two types, direct operated and pilot operated.



Direct operated relief valves “read” pressure and redirect fluid from the same inlet, and redirect fluid into the tank. The direct operated valve works by using a heavy spring which can be adjusted. Typically, these valves do not open immediately at the rated pressure, but instead crack open at about 50 to 60% of the full flow pressure.

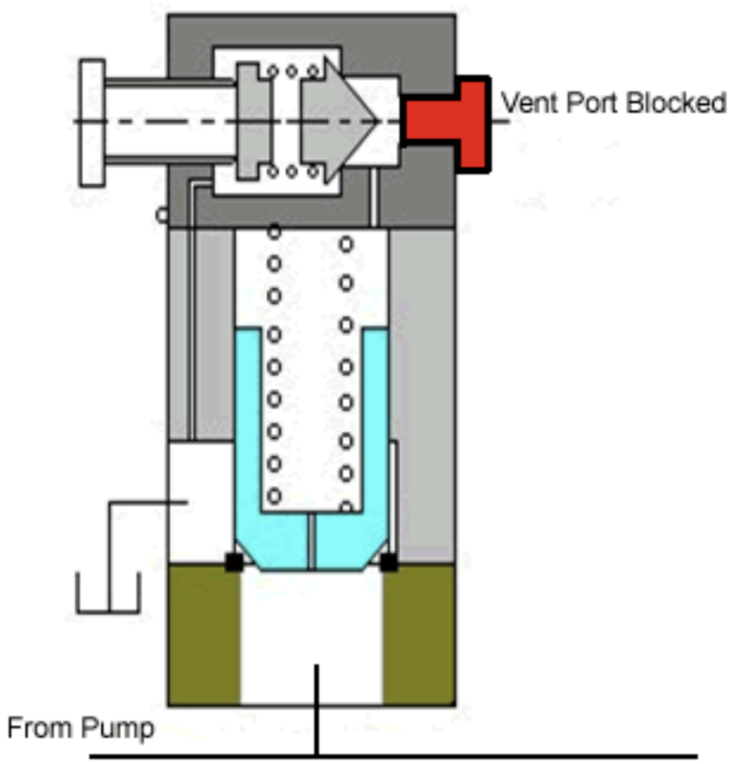


Fig. 7 Pilot Operated Relief Valve

Pilot operated valves “read” the pressure on a separate inlet to the redirected input, instead of using the same inlet. For relief valves, there are three pressure values at which the relief valve changes. When the fluid just escapes the valve, the pressure has reached the cracking pressure, which is said to be around 50 to 60% of the full flow pressure. Hence, the full flow pressure is the pressure where the valve fully opens and allows all of the fluid to flow through. The difference between the two pressure values is called the pressure override.

A pressure reducing valve instead reduces the pressure in a system, by limiting a branch’s pressure to be lower.

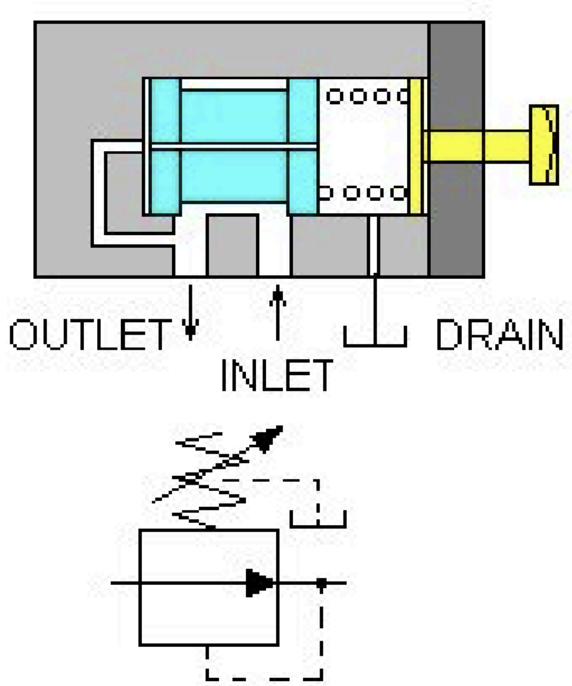
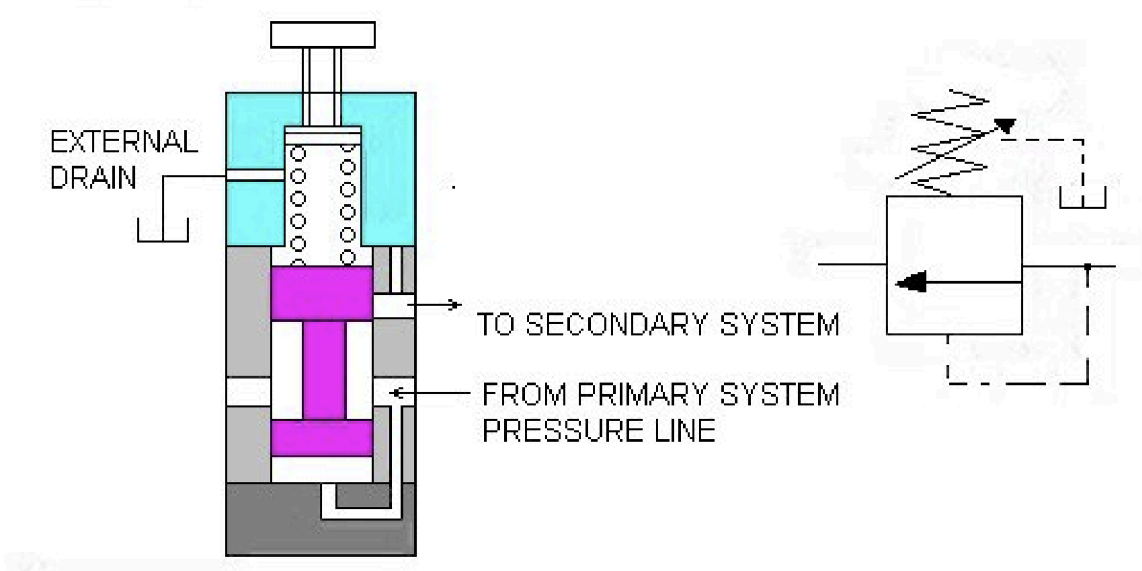
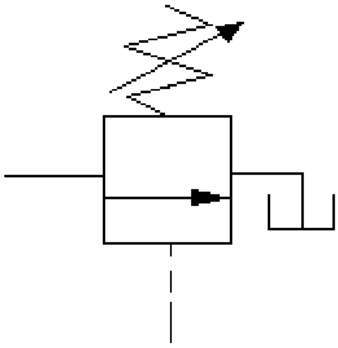


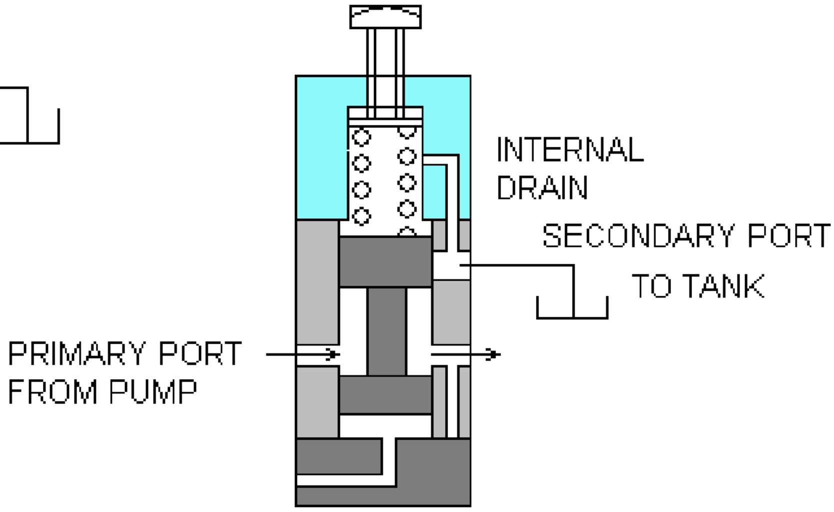
Fig. 10 Direct Acting Pressure Reducing Valve

Other than controlling pressure, valves can be used to sequence different hydraulic actuators, using pressure sequence valves, which allow a secondary system to activate once a primary system pressure has reached the pressure threshold.



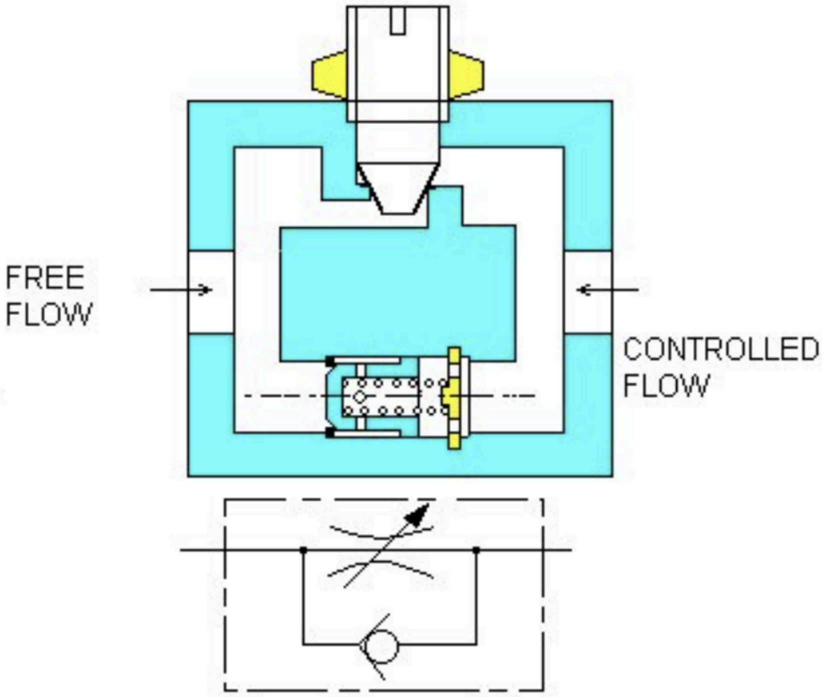
Another type of valve is an unloading valve, which allows fluid to flow back to the tank at zero pressure



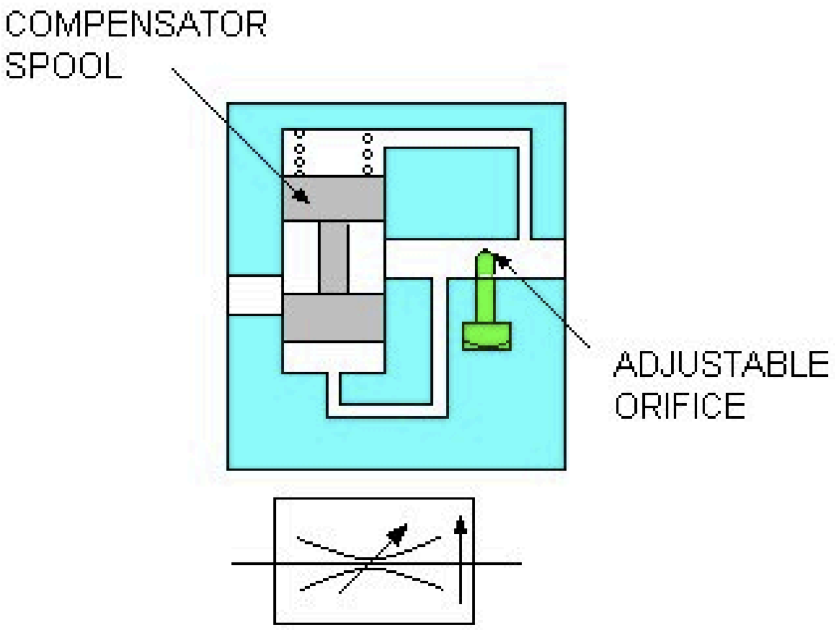


These pressure valves activate and deactivate based on the fluid's pressure. However, we often need valves that are externally controlled.

One example is a non compensated flow control valve, which throttles the flow in one direction but has a check valve to allow free flow in the other direction.

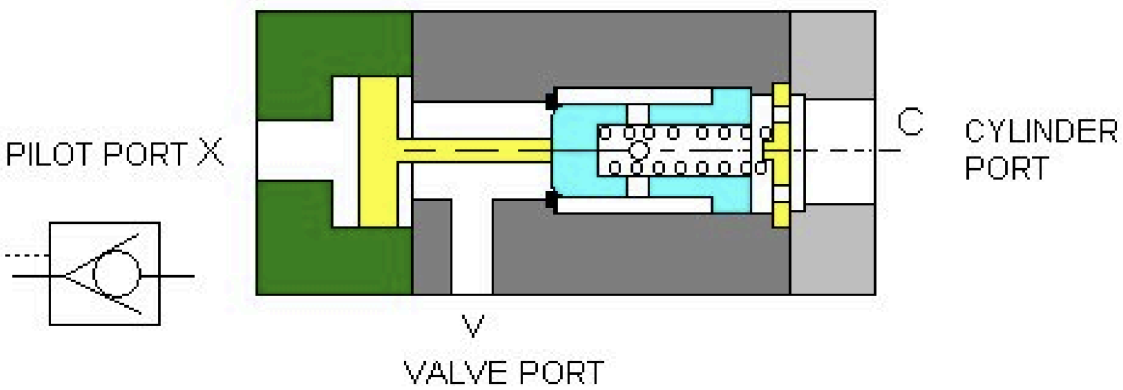


Pressure compensated flow control valves work by limiting the flow based on the pressure, by being normally open but closing up as pressure is applied, hence acting similarly to the flow control valve but being controlled by fluid pressure instead.



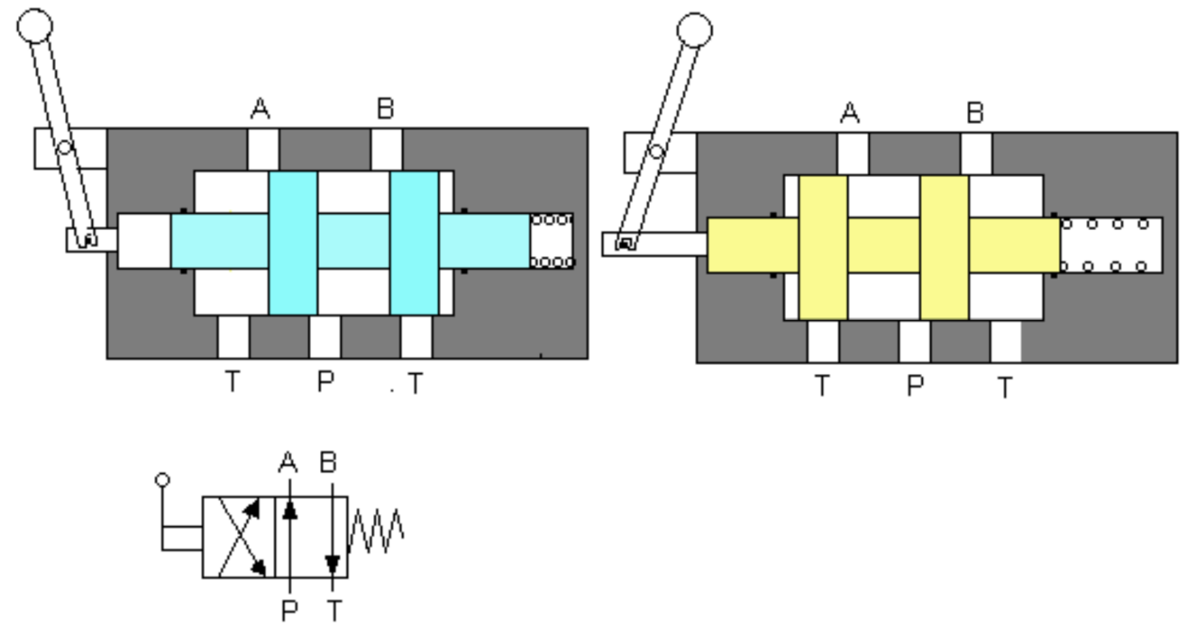
Another way to control valves is by using directional control valves; to redirect flow or only allow flow in one direction.

The most typical example is a check valve, but there is a different version called a pilot operated check valve, which only allows flow in one direction, but blocks flow in the other direction unless enough pressure is supplied into the pilot port.



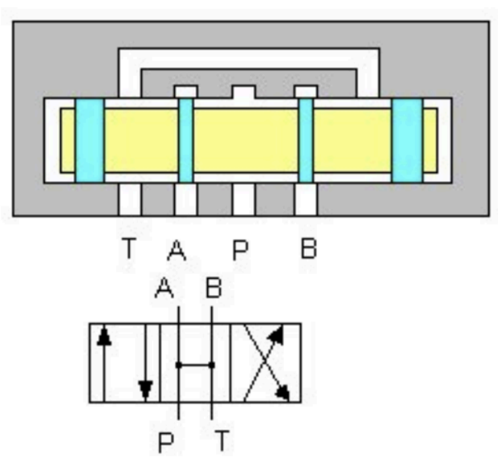
To manually control the fluid flow, we can use directional control valves.

Directional control valves are labelled by the number of ports, and the number of positions. For example, here is a 4/2 directional control valve



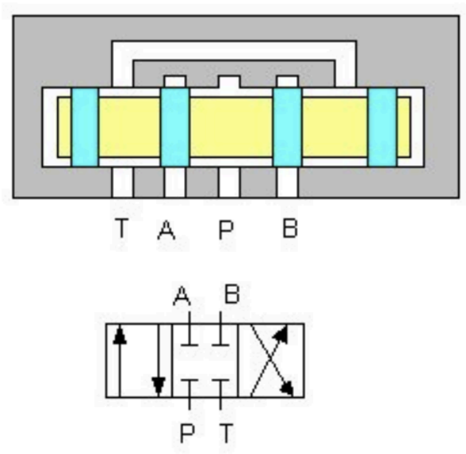
However, this type of control valve is only useful for 2 states, but 4/3 DCVs are more common, with a center position that acts as a neutral position. However, there are different ways the center position is arranged.

In open center spools, the ports are all connected to one another, used more for rotary actuators, as linear actuators would experience drift and not hold its position.



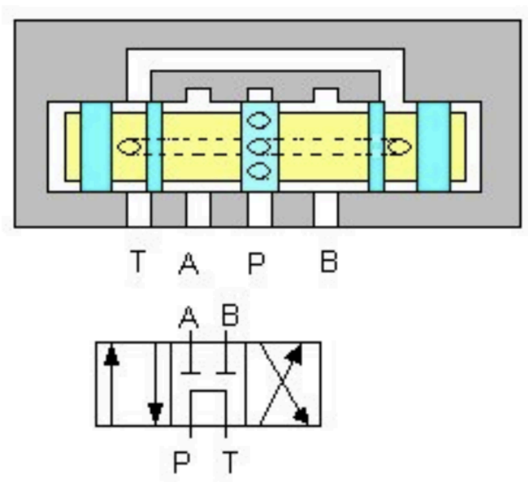
(a) Open Centre Spool

Close center spools cut every port to be isolated, hence requiring some other pressure relief system to relieve pressure



(b) Closed Centre Spool

A balance of the two is the tandem center spools, cutting A and B but leaving P and T connected, to provide a simple economical way to relieve pressure by letting fluid flow back into the tank.



(c) Tandem Centre Spool

The state the DCVs are in can be controlled manually with levers, electrically via solenoids, or pneumatically with pilot operated DCVs.

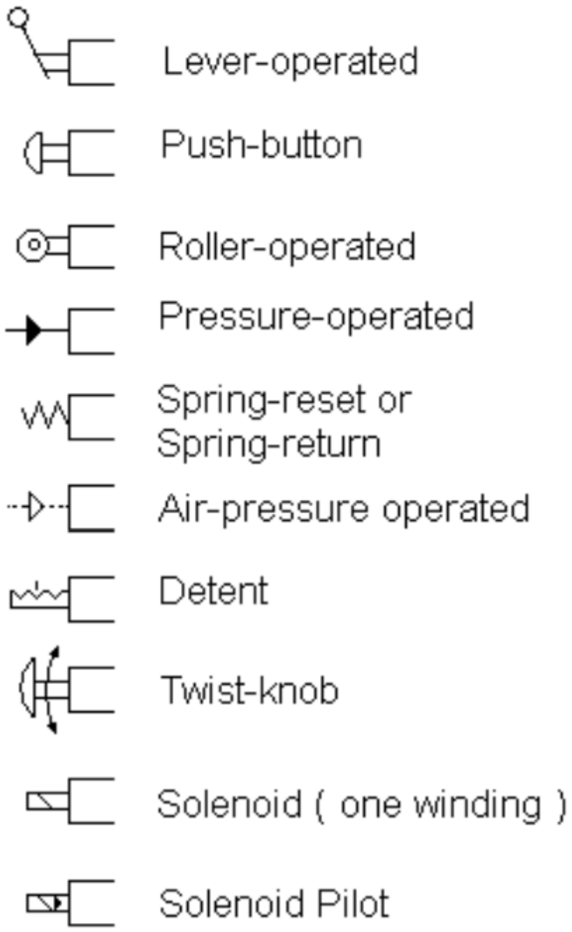


Fig. 20 Types Of Control

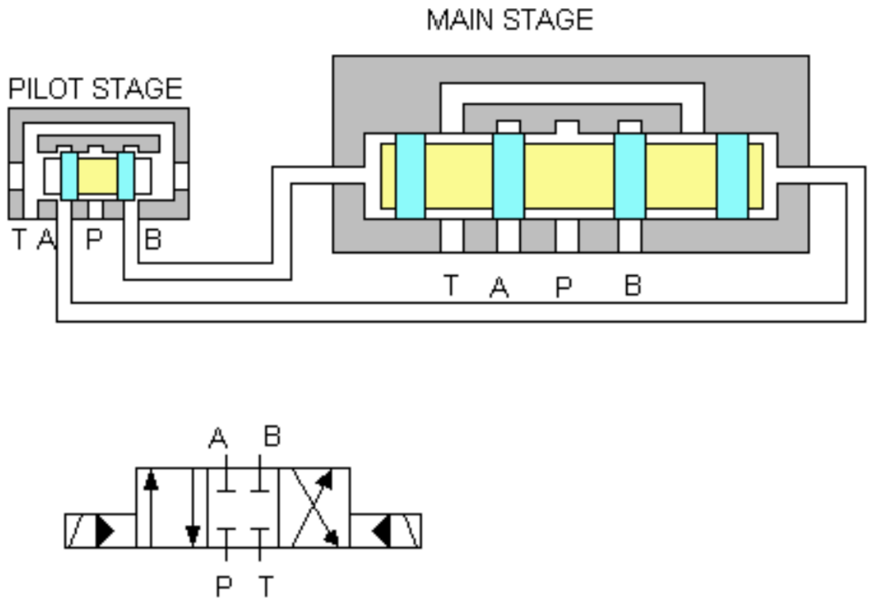
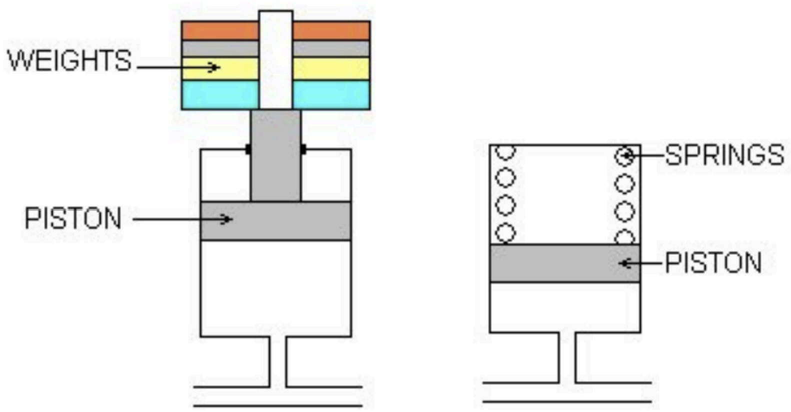


Fig . 19 Pilot Operated Directional Control Valve

In hydraulic circuits, accumulators may be added, to have a similar function to capacitors in electrical circuits. Accumulators store hydraulic fluid against dynamic force, and can supply said fluid when needed. There are three types, weight loaded, spring loaded and gas charged.



Gas loaded ones are more popular, having a gas reservoir that is able to compress when accumulating fluid and expand when depositing fluid.

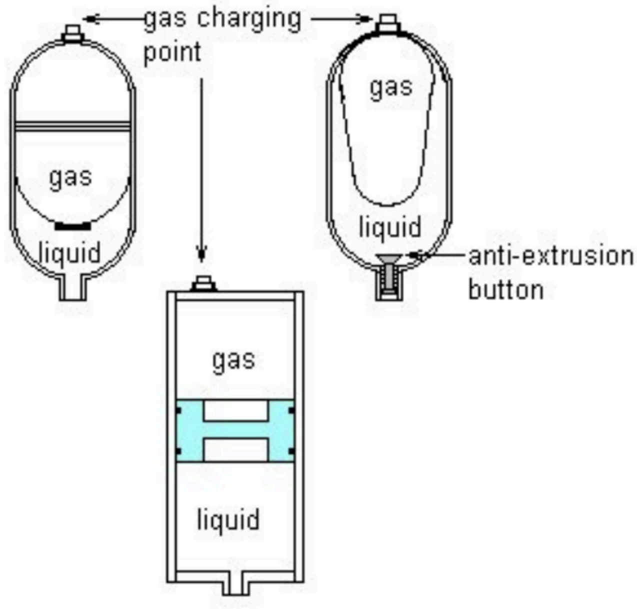
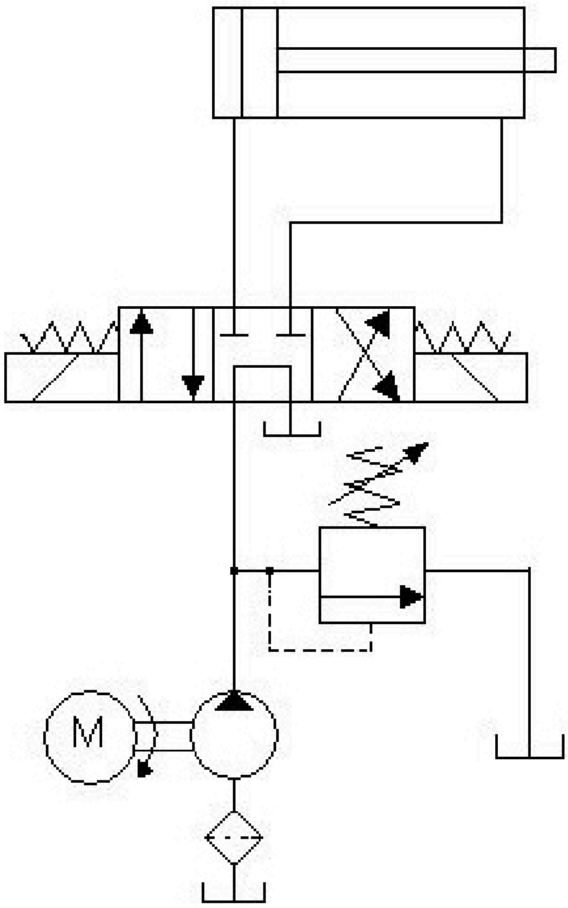


Fig. 25 Various Types Of Gas Charged Accumulator

Accumulators are able to store energy by acting as an energy reserve, absorb hydraulic shock, and help to compensate from pressure drops.

These hydraulic components are then used in hydraulic circuits to actuate cylinders in specific manners.

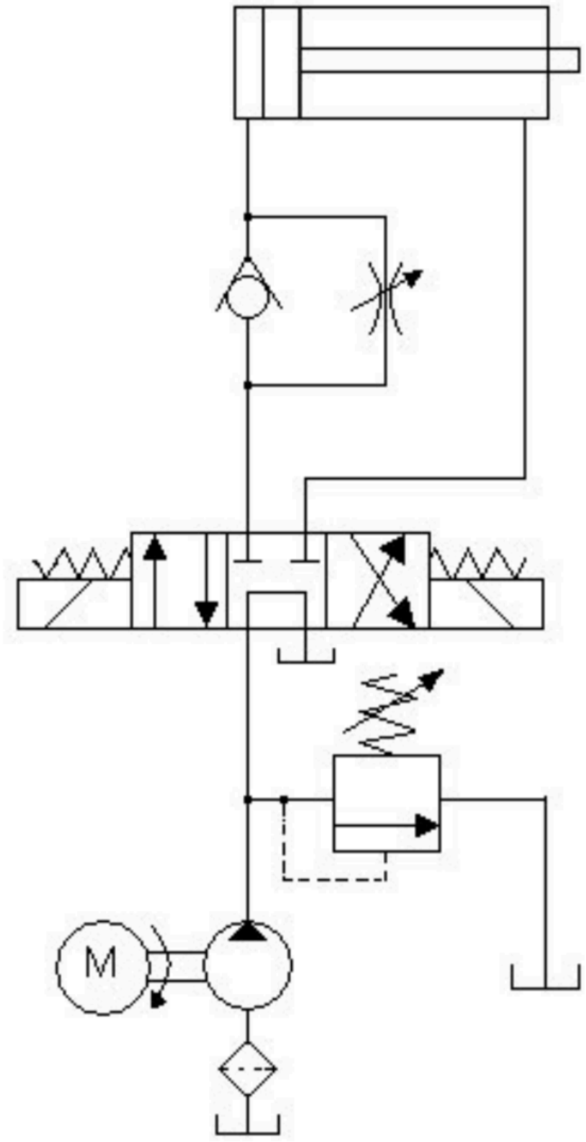
The typical conventional circuit is constructed like so.



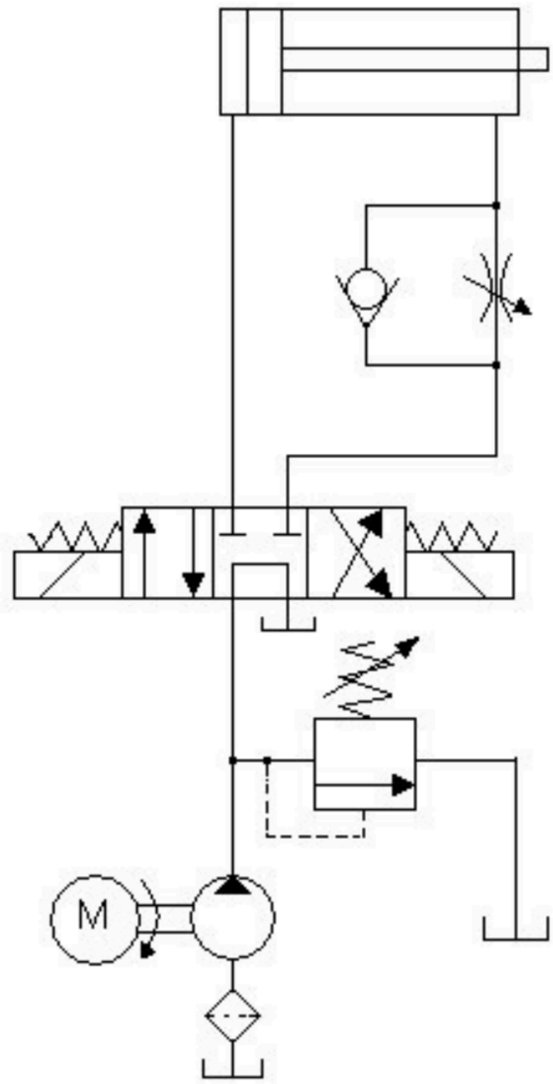
The conventional circuit takes fluid flowing from the power pack to directly do work on the cap end of the cylinder, pushing the piston while the fluid from the rod end goes into the tank. Additionally, a pressure relief valve is added to reject fluid if the pressure exceeds a certain limit.

One way to control the speed and force of the piston is by adding flow control valves before the cylinders inlets, giving the meter in and meter out circuits.

Meter in circuits have the flow control valve at the cap end of the cylinder, making it useful for slowing down the speed of the cylinder for constant resistive (positive) loads, like raising a load vertically, or pushing a load at a controlled speed like for grinding or milling. Hence, the meter in circuit control is also highly accurate too. Meter in circuits are not useful with unsteady loads, or negative loads (like when it has to pull something instead of push)



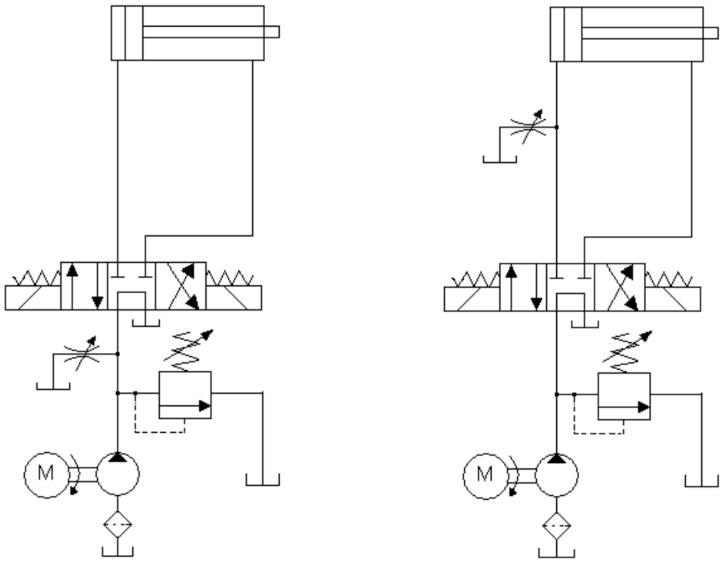
Meter out circuits have the flow control valve on the rod end of the cylinder instead:



This setup is more commonly used for drilling, reaming and boring, as the flow control valve controls fluid flowing out of the actuator, which makes it more stable under inconsistent loads (like from drilling), or more specifically, “runaway/overruning loads”. However, at higher pressures, a backpressure at the cap end may develop if the load becomes too erratic. Hence, relief valve protection needs to be done, at the cost of speed control.

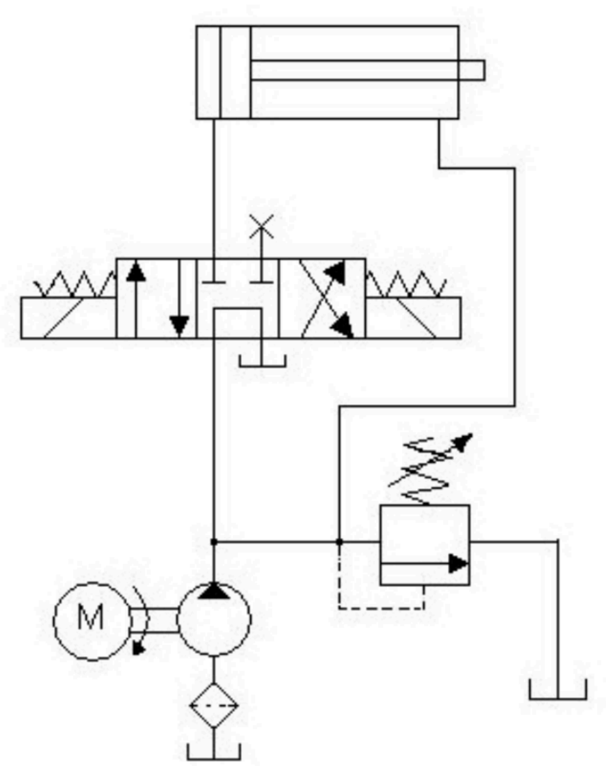
For both meter in and meter out circuits, as pressure develops as the flow is restricted, causing a lot of excess fluid to flow through the pressure relief valve and back into the tank, causing a significant loss in power.

One way to reduce the power loss from meter-in and meter-out circuits is by using bleed-off circuits.



Bleed off circuits have a flow control valve connected to the tank, to allow some fluid to bleed off from the main branch. This means that fluid in the cap end will accumulate slower than using a conventional circuit, requiring more pressure and flowrate to extend the cylinder. This requires less power as the pump only needs to operate at the pressure required by the cylinder. However, the control accuracy is lost compared to meter in and out circuits. Hence, this control is used in applications like large shapers or broaching machines where it can potentially consume a lot of power while also having a decent allowance for accuracy.

Note that these circuits are only able to slow down the piston, but cannot speed up the piston’s stroke. One method to speed up pistons is by using a regenerative circuit.



Regenerative circuits work by feeding the rod end of the cylinder back into the cap end of the cylinder, effectively increasing the flowrate. This causes the flowrate at the cap end to be the sum of the flowrate from the pump and the rod end, being

$$Q_T = Q_P + Q_R$$
$$vA = Q_P + v(A - a)$$
$$v\left(\frac{\pi}{4}D^2\right) = Q_P + v\left(\frac{\pi}{4}D^2 - \frac{\pi}{4}d^2\right)$$

where v is the steady state velocity of the extension. In the case where $A : a = 2 : 1$, then

$$vA : va = 2 : 1$$
$$Q_T : Q_R = 2 : 1$$
$$\therefore 2va = Q_p + va$$
$$Q_p = va$$

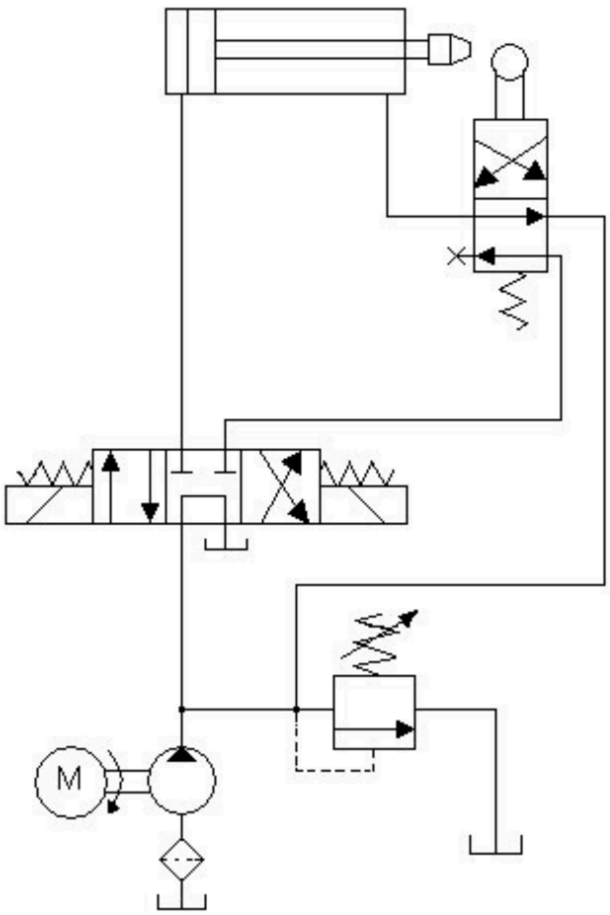
which means the speed of the piston is effectively doubled. However, towards the end of the stroke, the fluid at the piston end starts to get compressed, developing a backpressure at the piston end which resists the main force. The force at the end of the stroke would be

$$F = \frac{P_P}{A} - \frac{P_B}{A - a}$$

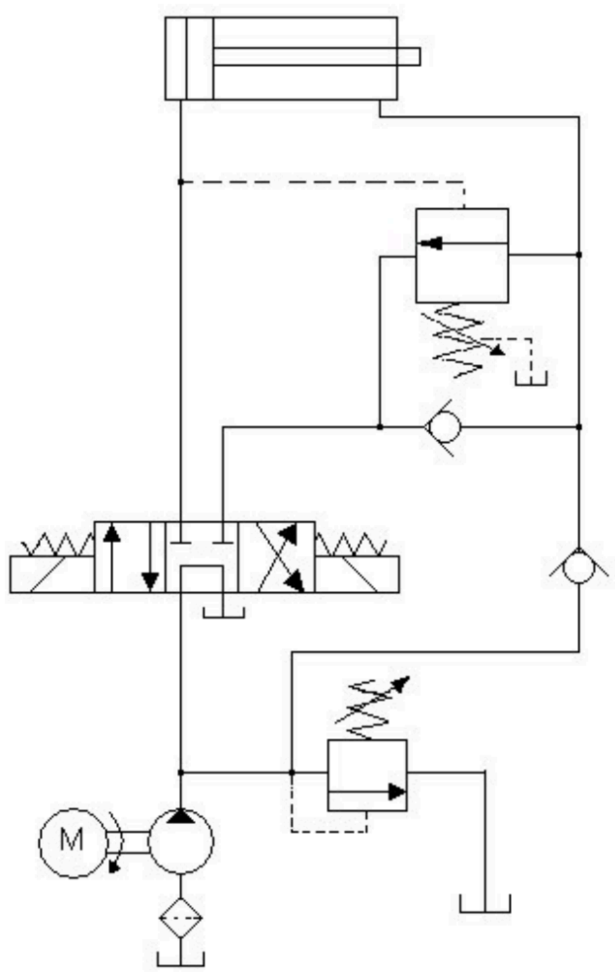
Hence, regenerative circuits have extra components added to switch the cylinder's extension mode from regenerative to conventional, to achieve the same force output while keeping the same speed.

Since the backpressure only develops at the end of the stroke, the regenerative circuit only needs to be switched to a conventional circuit at the end.

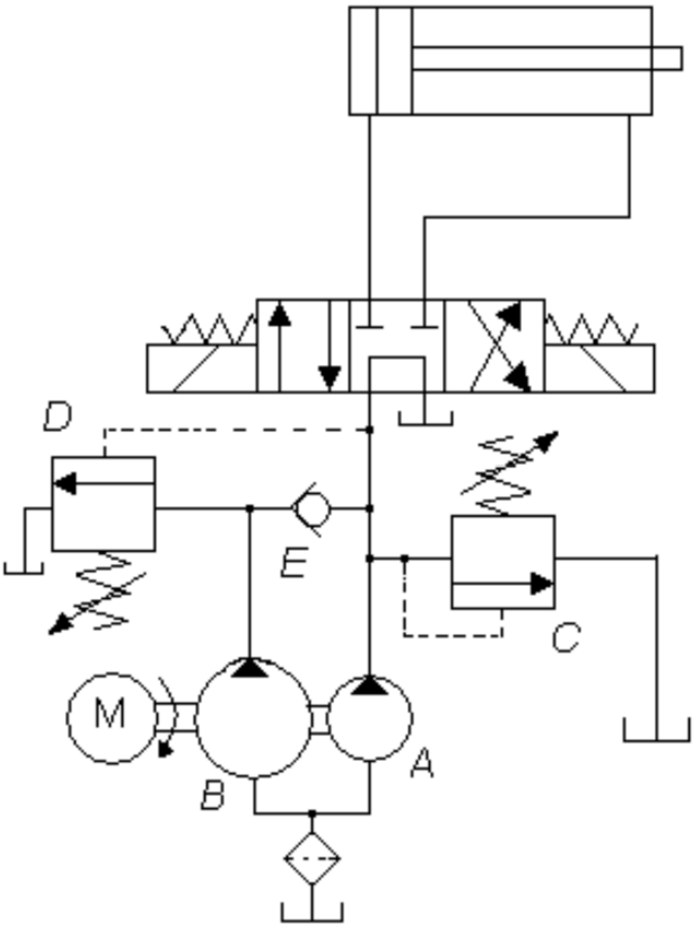
One method is by physically sensing when the piston has reached the end of the stroke, and using a directional valve to redirect flow from the rod end from going to the cap end to going into the tank.



However, another method is by sensing the pressure in the branch of the cap end, and once it starts to develop a pressure due to it resisting the backpressure developed at the rod end, it opens up another channel to allow for fluid from the rod end to flow back into the tank.

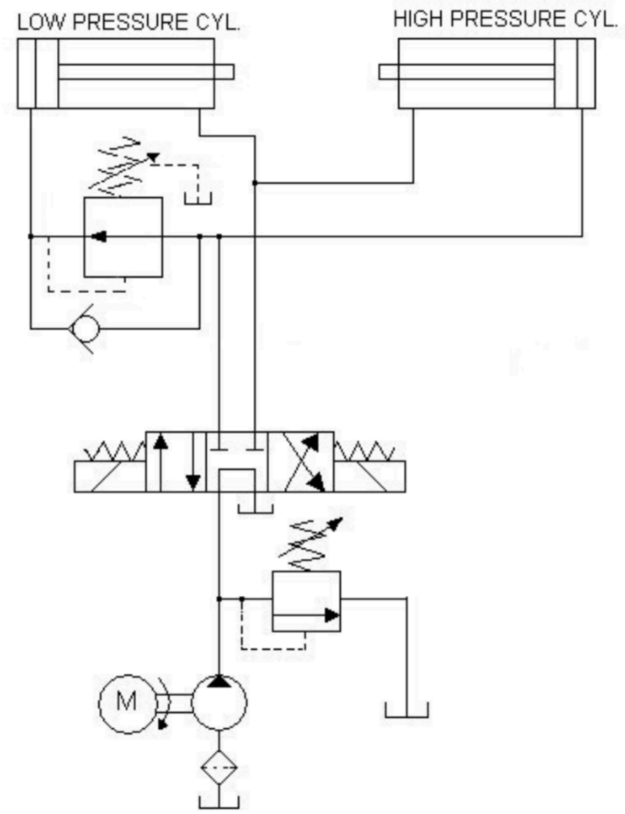


Another way to achieve high speed high force in a cylinder is by using two pumps in a Hi-Lo Circuit.



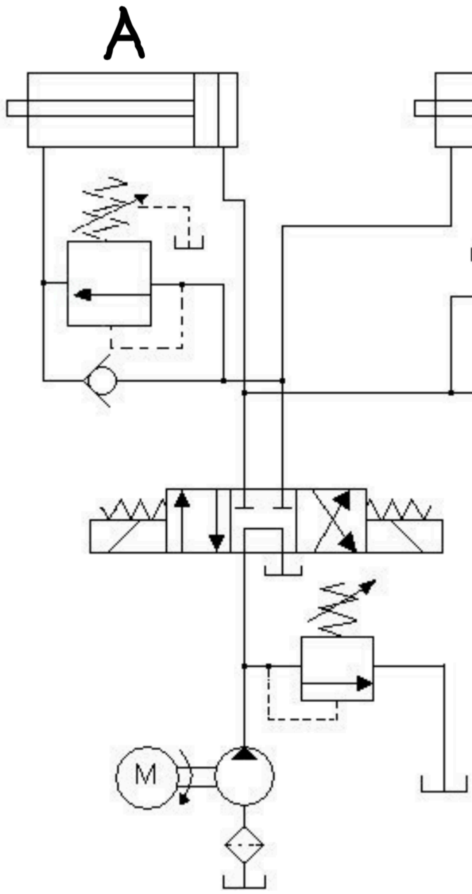
Two pumps are used, one of high flowrate but low pressure, and another of high pressure but low flowrate. During extension, fluid from both pumps goes into pushing the piston forward, hence the higher flowrate causes a quicker extension. However, as the cylinder reaches the end of the stroke, some backpressure developes which causes an increase in the pressure at the cap end branch. This activates the pressure sensing valve, which opens a branch for the high flow rate pump to flow into the tank, making it so that only the high pressure pump supplies fluid into the piston, producing a high force at the end.

Lastly for hydraulic circuits, there are circuits where two actuators can be controlled sequentially. One way is by using a pressure reducing valve.



When fluid originally flows into the DCV, fluid originally is only able to flow into the cap end of one of the cylinders, and once the cylinder fully extends, pressure develops in the branch, which allows the valve to open and the other cylinder to extend. When the DCV is reversed, both the rod ends of the cylinders are free to allow fluid to flow into it, allowing both cylinders to retract at the same time.

If we want one cylinder to retract before another, we can use two pressure reducing valves.



Like the earlier circuit, the cylinder connected to the pressure reducing valve extends last, hence A extends first, then B. During retraction, fluid flows immediately into the rod end of cylinder B, allowing B to retract, before activating the valve to retract A.